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Preferential Leaching of Water and Chloride in a Clay Loam Soil as Affected by Tillage and Rainfall Intensity

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There is a concern that preferential flow (i.e., bypass of soil matrix) of water and chemicals through soil macropores may enhance leaching below the root zone of agricultural crops, which may impact groundwater quality. Our objective was to investigate the nature of preferential flow in a clay loam soil and to ascertain the potential impact of tillage practice and rainfall intensity. We utilized a long-term (since 1968) conventional tillage (heavy-duty cultivator) and no-till field located at Lethbridge in southern Alberta. Four large (46 × 46 × 51 cm) undisturbed soil blocks were excavated from each tillage field (unreplicated) during the fallow phase of a wheat-fallow rotation. The soil blocks were transported to the laboratory, stored under drying conditions (32 months), and rainfall-leaching experiments conducted under unsaturated, transient conditions. In the laboratory, we attached a funnel-lysimeter consisting of 64 square funnels or cells to the bottom of each block. Three rainfall experiments were conducted on each block and a solid chloride tracer was applied to the top of the soil block before each rainfall event. Blocks were at a soil water content of field capacity prior to each rainfall. A total of 22 mm of rainfall was applied in 60, 30 and 15 min, corresponding to rainfall intensities of 22, 44 and 88 mm h^{-1}, respectively. The volume of water outflow and concentration of chloride in each cell was determined after each rainfall event. Water outflow through ≥98% of cells indicated that many soil pores contributed to leaching of water. Convex-shaped curves of cumulative water outflow from cells versus cumulative area indicated non-uniform water flow, but the degree of water flow preference was low. Semi-variograms of cell water outflow indicated random flow, and the pattern of cell water outflow was stable with time and among rainfall intensities. Chloride outflow in ≤14% of cells indicated that only a few soil pores contributed to leaching of chloride. Total mass loss of chloride from each soil block was <0.1% of the total amount applied, which suggested most chloride was retained within the soil block, and was more consistent with immobile water rather than anion exclusion. Tillage practice seemed to exert a stronger influence on water outflow parameters than rainfall intensity, with a trend towards greater preferential flow of water under CT than NT. Total block water outflow increased with greater rainfall intensity for CT, and to a lesser extent for NT. Mean values of maximum chloride concentration were higher for CT than NT at 88 mm h^{-1}, but not at the two lower intensities. Rainfall intensity had no effect on mean values of maximum chloride concentration for either CT or NT. Our results indicated that although preferential leaching occurred in these wet clay loam soils under relatively high rainfall intensities, the amount of chloride leached past 50 cm was minimal. Caution is advised in extrapolating these results to scenarios where conditions are different from this study.

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Introduction

Preferential flow (i.e., bypass of soil matrix) of water and solutes through soil macropores is a major factor affecting leaching below the root zone and groundwater quality (Bouma 1991; Bevan 1991). If we understand the mechanism or process of preferential flow in soils, and how it is affected by rainfall intensity and tillage practices, we can manage our soils using agricultural practices (e.g., tillage, irrigation, crop type and rotations) to minimize leaching below the root zone.

Methods for measuring preferential flow have been reviewed by (Bouma 1991), Luxmoore (1991) and Edwards et al. (1993). A simple method for measuring preferential flow consists of excavating undisturbed soil blocks in the field and transporting them to the laboratory (Andreini and Steenhuis 1990). A funnel-lysimeter is then attached to the bottom of the block to measure preferential flow. The funnel-lysimeter consists of a grid of square funnels (equal size) or cells. A chemical tracer is usually applied to the soil surface and leached through using a rainfall simulator. If all cells collect similar volumes of percolate and concentrations of the tracer, this indicates uniform or piston plow; but if only a few cells collect percolate, this indicates nonuniform or preferential flow. The grid network of cells also facilitates investigation of the spatial variation of water and solute flow patterns.

The soil block method has been used to study preferential flow in soils of New York (Andreini and Steenhuis 1990), Kentucky (Quisenberry et al. 1994; Phillips et al. 1995), Ohio (Shipitalo et al. 1990; Edwards et al. 1992; Granovsky et al. 1993; 1994), Ontario (Bowman et al. 1994) and Denmark (Wildenschild et al. 1994). The studies used rainfall intensities ranging from 0.83 to 120 mm h^{-1}. Water flow (steady-state or transient), soil water conditions (unsaturated or saturated) and lower (negative or zero soil water potential) and upper (exposed soil or protective sand layer) boundary conditions were variable among the studies. Most soil block studies have observed nonuniform water and chemical flow through the funnel-lysimeter, which indicated preferential flow in macropores. However, the degree of nonuniformity varied among studies.

Soil block studies have focused on the mechanism of preferential flow and the impact of management practices such as tillage and residue cover. A negative relationship was observed between antecedent soil water content and water outflow in no-till (NT) soils, and pre-rain water content was found to be the major factor affecting transport (Granovsky et al. 1994). Most researchers have reported increased water and chemical outflow with increased rainfall intensity (Edwards et al. 1992; Quisenberry et al. 1994). However, Phillips et al. (1995) found that outflow of water and chloride through macropores increased as the application rate decreased.

The sequence of rainfalls of different intensities can also affect leaching. If the first rainstorm after chemical application is of low intensity, the potential for leaching in subsequent storms of higher intensity is reduced (Shipitalo et al. 1990; Edwards et al. 1992). However, if the first rainstorm after chemical application is of high intensity, there is the potential to leach chemicals below the root zone. Some studies have found that rainfall timing and intensity following chemical application may exert a stronger influence on leaching than management practices such as tillage (Granovsky et al. 1993). Researchers have reported both stable (Quisenberry et al. 1994) and unstable (Wildenschild et al. 1994; Ogden et al. 1999) water outflow patterns among consecutive rainfall events, which may be related to the presence or absence of a protective sand layer on top of the soil blocks. A protective sand layer prevents surface sealing of macropores (Ela et al. 1992), which contributes to stable outflow patterns through a funnel-lysimeter.

Although studies have investigated preferential leaching in no-till soil blocks (Shipitalo et al. 1990; Edwards et al. 1992; Granovsky et al. 1994), we are aware of only two studies (Andreini and Steenhuis 1990; Granovsky et al. 1993) that have compared water and chemical outflow through soil blocks from conventional tillage (CT) and no-till (NT) fields. Andreini and Steenhuis (1990) found that there was little difference in water or chemical transport by preferential flow between CT and NT tillage practices. They attributed this to the low water application rate employed and the timing of soil block excavation in relation to the last tillage event. Granovsky et al. (1993) reported that NT blocks transmitted greater water and chemical amounts than CT (moldboard plow); however, the differences were rarely significant. They also found that hydraulic behaviour was affected by the interaction of soil type and tillage treatment. For example, preferential flow was greater for CT than NT for a Hoytville silty clay loam, but was greater for NT than CT for a Crosby silty loam.

We are unaware of any researchers who have used the soil block method to compare water and chemical outflow in CT and NT soils of the Northern Great Plains. Of the soil block studies reviewed, only one of the studies was conducted on heavier textured soils (most were done on fine sandy loam to silt loams) and none of the studies were done on soils cropped to cereals (crops grown were corn, bluegrass or alfalfa). The objective of this study was to utilize the soil block method and investigate the nature and distribution of preferential flow in clay loam soils of southern Alberta. A secondary objective was to examine the effect of tillage practice and rainfall intensity on preferential flow.

Materials and Methods

Undisturbed soil blocks ($46 \times 46 \times 51$ cm) were excavated and collected from long-term (since 1968) tillage fields (non-replicated) on a Dark Brown Chernozemic clay loam soil at Lethbridge in 1992. Details of the study soil, excavation and soil block preparation have been previously reported (Miller et al. 1999a). Four soil blocks were collected from a conventional tillage (CT) field and four from a no-till (NT) field. Conventional tillage was performed with a chisel plow or heavy-duty cultivator and rod-weeder attachment to a depth of approximately 10 cm. No-tillage used only herbicides for weed control. The sampled plots were in the fallow phase of a 2-year wheat (*Triticum aestivum L.*)-fallow rotation.

Soil blocks were transported to the laboratory at the Lethbridge Research Centre and stored for 32 months before being tested. The soil blocks were sealed and stored at 20°C and no water was added during the storage period to minimize biological activity. We estimated the soil water content to be between 10 and 20% during the storage period. The soil blocks were checked once a week for the first month, and then every month, to ascertain the soils condition in terms of earthworm activity, microbial activity (e.g., mould, algae growth), extent of soil cracking and soil shrinkage. Earthworm activity, as evidenced by casts on the soil surface, appeared to cease within the first month of storage. We observed little or no evidence of moulds on the soil surface during the storage period, which was consistent with the increasingly dry soil water conditions. There was only minor soil cracking during storage, and these cracks swelled shut and disappeared after saturation. There was also no visual evidence of soil shrinkage and soil pulling away from the foam liner. Although these soils are dominated by smectite clays (Kodama 1979), the shrink-swell potential, as indicated by coefficient of linear extensibility (COLE) values (0.06), was only moderate (Schafer and Singer 1976). Leaching experiments were initiated in May, 1995, and were completed in May of 1996.

In the laboratory, a funnel-lysimeter used to measure preferential flow was attached to the bottom of the soil block before testing. The funnel-lysimeter and experimental setup are shown in Fig. 1. This apparatus consisted of 64 square (5 × 5 × 4 cm) funnels (stainless steel) in the centre, surrounded by four edge-trays (Fig. 1a and b). The soil block was mounted on a stand and the funnel-lysimeter attached on the bottom (Fig. 1c). A fiberglass sheet was laid on top of the soil block and about 1 to 2 cm of fine sand (0.13 mm diam.) was added to achieve a level surface. The purpose of the sand was to minimize disruption of the soil surface and improve uniform delivery of water to the soil surface. Protective sand layers are commonly used in solute transport experiments (Quisenberry et al. 1994) and should have no effect on infiltration into fine-textured soils, since the layer of least permeability dominates water flow through layered systems (Jury et al. 1991).

Leaching experiments were conducted immediately after previous experiments (Miller et al. 1999a) where the soil was saturated and then allowed to drain by gravity until no water outflow occurred (i.e., field capacity). We added a chloride tracer (as solid KCl) to the top of the sand, applied a designated rainfall intensity, and then measured the volume of water and concentration of chloride leached through the 64 funnels on the bottom of each soil block. The bottom of the soil block was kept open to

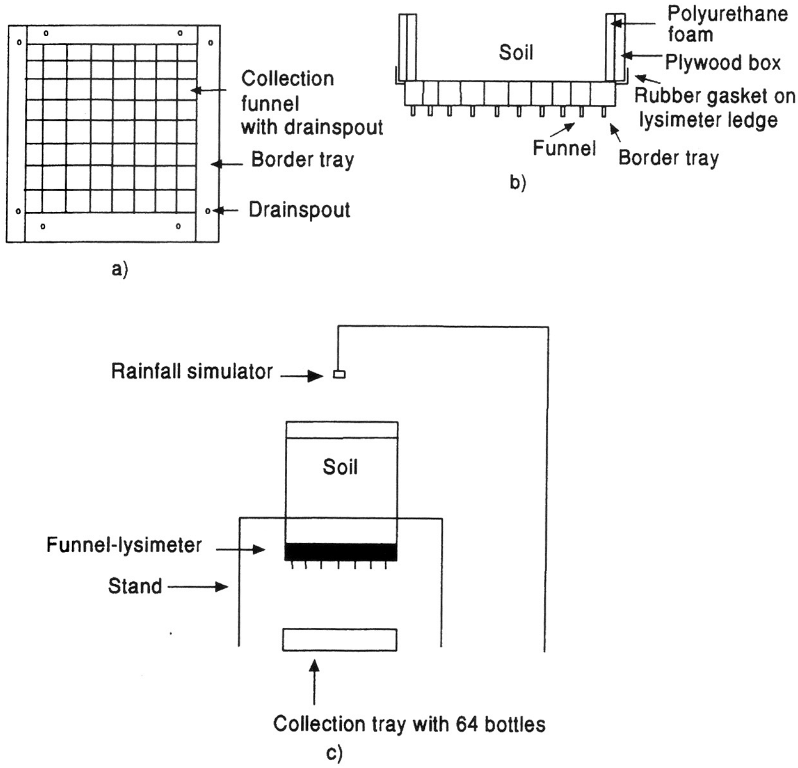

Fig. 1. Top-view (a) and side-view (b) of funnel-lysimeter used to collect effluent and experimental set-up (c) for conducting laboratory leaching experiments.

the atmosphere (soil water potential of zero) and effluent was allowed to drain by gravity. This bottom boundary condition is commonly used to study preferential leaching in macropores (Andreini and Steenhuis 1990; Shipitalo et al. 1990; Wildenschild et al. 1994). Since leaching of water and solutes increase with an increase in soil water potential at the lower boundary (Phillips et al. 1995), gravity drainage at zero water potential represents a worst case scenario for preferential leaching. A tray consisting of 64 square plastic bottles (500 mL) was used to collect the percolate. A solid chloride tracer (4430 mg of Cl^- as KCl) was added to the top of the sand layer by weighing out 64 equal portions of the tracer and placing each portion above the location of each funnel. The total amount of chloride applied was equivalent, in terms of moles per unit area (0.50 moles Cl^- or NO_3^- per m^2) to the amount of NO_3^- in a 130 kg N ha^{-1} application of NH_4NO_3. The chloride tracer was added prior to applying each rainfall intensity.

A Guelph rainfall simulator (Tossel et al. 1987) was used to apply rainfall intensities of 22, 44 and 88 mm h^{-1} in 60, 30 and 15 min, respectively. These rainfall intensities are equivalent to annual extreme values for return periods of 2 to 5, 5 and 10 yr, respectively (Tautchin et al. 1991). Uniformity of rainfall application, often referred to as the uniformity coefficient (UC), was determined using the method of Christiansen (1942). The collection tray containing the 64 square bottles was used for the uniformity tests and five replications were conducted for each rainfall intensity. The mean UC values were 92, 94 and 93% for the low, medium and high rainfall intensities, respectively.

The same soil block was used for each rainfall intensity experiment. Rainfall experiments were conducted in the sequence from lowest to highest intensity, and the chloride tracer was added before each rainfall intensity experiment. Soil water content before each rainfall intensity experiment was determined at three depths (10, 25, 45 cm) using time-domain reflectometry (Topp 1993), and as described in detail by Miller et al. (1999a). The soil block was leached with a minimum of four pore-volumes of chloride-free water before the next rainfall intensity was applied. After application of each level of rainfall intensity, the volume of percolate in each of the 64 botttles was weighed (assuming density of 1 g cm^{-3}) and a subsample taken for chloride analysis. Percolate volumes are expressed as a percentage of one pore volume because soil block volume and soil porosity varied among blocks. The value of one pore volume was calculated by multiplying total porosity by the total volume of the soil block. Chloride concentration was determined using the mercury thiocyanate-autoanalyzer method (Technicon 1974). The detection limit for chloride was 2.13 mg L^{-1}. The number of visible (>500 μm diameter) pores on the bottom of each soil block was counted immediately before the tracer test was conducted on each soil block.

The degree of water flow preference was estimated by sorting cell percolate volumes in decreasing order and then plotting the cumulative relative outflow against the cumulative relative area (Granovsky et al. 1994). If all 64 cells contributed equally to outflow (i.e., no preferential flow), the plot would be a straight line. A non-uniform distribution of cell percolate volumes (i.e., preferential flow) would generate a convex upward curve. We tried fitting an exponential function used by Granovsky et al. (1994) to the outflow curves to obtain a numerical estimate of the degree of flow preference (i.e., slope), but found this method inadequate because of the different curve shapes and large errors associated with fitting the equation to the curves. Consequently, we devised a simpler method that we feel gives a more accurate estimate of the degree of water flow preference. We measured the area between the 1:1 line and the relative outflow curve using a graphics software package. Completely uniform flow described by the 1:1 line gave an area of zero. The highest degree of flow preference would have 100% of the water outflow occurring through 1/64 (0.02) of the relative area (i.e., one cell only). This extreme preferential flow would have a vertically ascending line to a relative outflow value of 1.0 (y value) from a relative area value of 0.02 (x value). For this curve, the area between the 1:1 line and the curve would be close to 0.50. Therefore, the degree of water flow preference can vary from nonexistent (area=0) to extreme (area≈0.50).

The spatial variation of water outflow (in units of milliliters) was investigated by constructing semi-variograms using the Geostatistical Environmental Assessment Software, GEO-EAS (Englund and Sparks 1988). If required, values for water outflow were log-transformed to attain a normal distribution prior to running GEO-EAS. Since flow was found to be isotropic, the direction for the pair orientation was set at 0°(parallel to x axis) and the directional tolerance was set at 90°. Only ≥30 lag pairs were used in constructing the semi-variogram.

In this experiment, there was no replication of the CT and NT tillage treatments. Consequently, means and standard errors (SE) are reported for tillage and rainfall intensity comparisons to indicate overall trends, but statistical probability levels are not included. We determined the mean±SE for comparisons of interest and, if they did not overlap, this was taken as a trend towards a treatment difference.

Results and Discussion

Water Outflow

Cell outflow, expressed as a percentage of one pore volume, was variable within each soil block, among soil blocks from the same tillage field, between tillage fields and among rainfall intensities (Fig. 2 and 3). The nonuniform outflow within each soil block indicated preferential flow. All 64 cells conducted water for the soil blocks under the experimental conditions in this study; with the exception of blocks CT1 and CT3 at the low rainfall intensity, where 63 cells conducted water. Outflow through ≥98% of cells indicated that leaching of water occurred through many soil pores. In contrast, Edwards et al. (1992) reported a much lower number of cells (0 to 27% of cells) conducting water, which indicated fewer soil pores contributing to leaching.

Total water outflow for blocks as a percentage of one pore volume ranged from 6 to 11% (Table 1), and as a percentage of the amount applied it ranged from 68 to 122% (data not shown). Mean values for both parameters were higher for CT than NT blocks at 22 mm h^{-1}, and total water outflow as a percentage of one pore volume generally increased with greater rainfall intensity. In contrast, Ogden et al. (1999) found that total block outflow was significantly greater for NT than PT (plow till), and Granovsky et al. (1993) reported a similar but non-significant trend. Edwards et al. (1992) also reported a positive relationship between block water outflow and rainfall intensity.

The CV values for outflow ranged from 45 to 105% (Table 1), reflecting a wide range in outflow through small to large pores. Mean CV values were generally greater for CT than NT at all intensities, but rain intensity had no impact on this parameter. Maximum values for cell water outflow as a percentage of one pore volume ranged from 0.23 to 0.69% (Table 1), and as a percentage of the total block outflow they ranged from 3.3 to

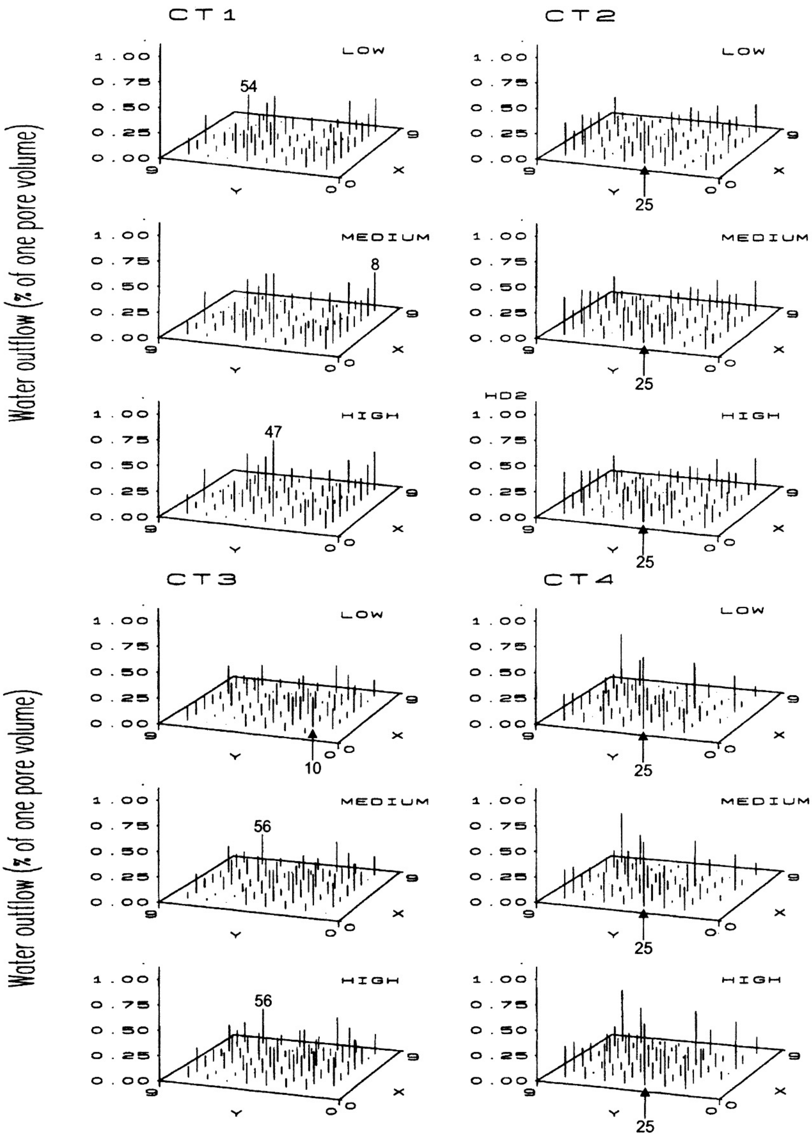

Fig. 2. Cell water outflow (% of one pore volume) from soil blocks taken from conventional tillage (CT) field in southern Alberta. Rainfall intensity treatments are low, medium and high intensities (22, 44 and 88 mm h^{-1}, respectively). Cell with maximum water outflow is indicated on graph.

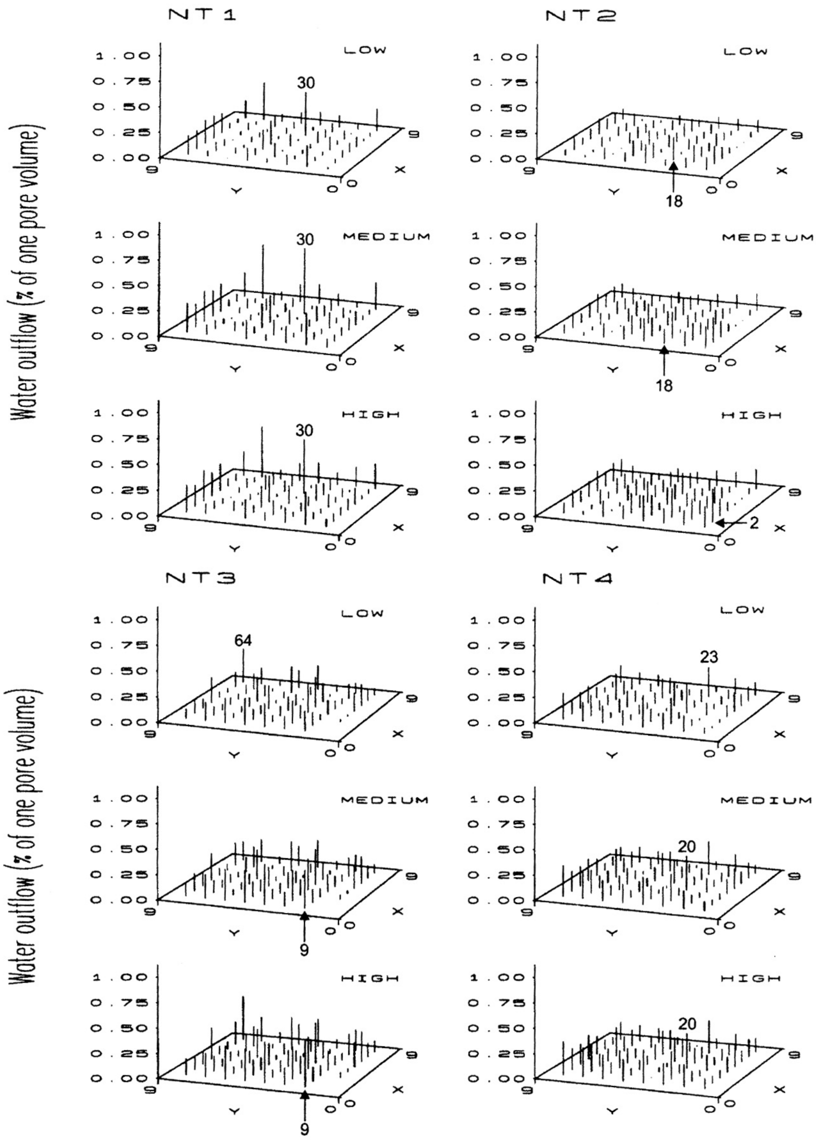

Fig. 3. Cell water outflow (% of one pore volume) from soil blocks taken from no-till (NT) field in southern Alberta. Rainfall intensity treatments are low, medium and high intensities (22, 44 and 88 mm h^{-1}, respectively). Cell with maximum water outflow is indicated on graph.

Table 1. Selected soil and water flow parameters for soil blocks

Rainfall intensity mm h^{-1}	CT1[a]	CT2	CT3	CT4	Mean	SE	NT1[b]	NT2	NT3	NT4	Mean	SE
	Antecedent soil water content for block (% by volume)											
22	34.3	40.0	35.2	35.1	36.2	1.3	43.3	41.5	39.6	41.8	41.6	0.8
44	34.5	38.8	40.2	35.2	37.2	1.4	43.0	41.9	40.0	41.9	41.7	0.6
88	35.2	40.0	35.2	21.9	33.1	3.9	44.3	42.7	40.0	41.4	42.1	0.9
	Total water outflow for block (% of one pore vol.)											
22	8.3	8.6	7.8	8.4	8.3	0.2	6.3	6.1	8.1	7.9	7.1	0.5
44	8.7	9.7	8.6	8.6	8.9	0.3	8.5	7.9	9.9	9.4	8.9	0.5
88	9.1	9.4	10.0	11.0	9.9	0.4	8.4	8.8	11.0	10.0	9.6	0.6
	Degree of water flow preference for block (dimensionless)											
22	0.23	0.19	0.20	0.27	0.22	0.02	0.22	0.13	0.16	0.18	0.17	0.02
44	0.24	0.18	0.19	0.27	0.22	0.02	0.21	0.13	0.17	0.14	0.16	0.02
88	0.24	0.18	0.24	0.24	0.23	0.02	0.20	0.12	0.17	0.17	0.17	0.02

(continued)

Table 1. *(concluded)*

Rainfall intensity mm h^{-1}	CT1[a]	CT2	CT3	CT4	Mean	SE	NT1[b]	NT2	NT3	NT4	Mean	SE
	Coefficient of variation for cell water outflows (%)											
22	83	70	70	105	82	8	88	49	66	58	65	9
44	85	65	68	103	80	9	91	46	62	53	63	10
88	86	67	61	89	76	7	82	45	63	52	60	8
	Maximum cell water outflow (% of one pore volume)											
22	0.37	0.42	0.29	0.69	0.44	0.09	0.44	0.23	0.35	0.30	0.33	0.04
44	0.40	0.43	0.31	0.63	0.44	0.07	0.67	0.26	0.37	0.35	0.41	0.09
88	0.47	0.43	0.36	0.60	0.47	0.05	0.54	0.32	0.47	0.36	0.42	0.05

[a]CT, conventional tillage. Values for one pore volume for CT blocks were 51,398 (CT1), 48,904 (CT2), 46,758 (CT3) and 48,934 (CT4) cm^3.
[b]NT, no-till. Values for one pore volume for NT blocks were 57,481 (NT1), 53,753 (NT2), 49,197 (NT3) and 53,947(NT4) cm^3.

8.2% (data not shown). Edwards et al. (1992) reported much higher values (34 to 71%) for the latter parameter, which indicated greater preferential flow in their soils. Maximum cell water outflow (as percentage of one pore volume) was greater for CT than NT at 22 mm h^{-1}, and there was little or no effect of rain intensity on this parameter.

Semi-variograms for cell outflow generally revealed random flow with no spatial component for lag distances from 6 to 39 cm (Fig. 4). The only exceptions were blocks CT2 and CT4, where a weak spatial component was detected. Rainfall intensity generally had no effect on the semi-variance. Similar findings were reported for cell outflow from soil blocks taken from CT (moldboard plow) and NT soil plots in Ohio (Granovsky et al. 1993).

The degree of water flow preference for each soil block was assessed by plotting cumulative relative cell outflow versus cumulative relative area (Fig. 5). The gently rising convex curves above the 1:1 line (uniform flow with each cell collecting same volume) indicated a low degree of flow preference. Quisenberry et al. (1994) reported similar flow preference curves for a silt loam soil under bluegrass from Kentucky. In contrast, Granovsky et al. (1994) found a high degree of flow preference for NT soils in Ohio containing the earthworm *Lumbricus terrestris* (Edwards et al. 1992). Their flow preference curves had very steeply rising limbs and 100% of the water outflow was conducted through between 10 to 60% of the area, indicating extreme flow preference in some soil blocks. Bowman et al. (1994) also reported a high degree of flow preference in silt loam soils from Ontario, where more than 99% of the water flow was conducted through only about 26% of the area.

The degree of water flow preference in our soil blocks was clearly higher for CT than NT at all three intensities, but rainfall intensity had no effect on either the CT or NT blocks (Table 1). The higher degree of water flow preference for CT than NT may be due to the greater rooting density in this CT field (Volkmar and Entz 1995). Miller et al. (1999b) also reported higher unsaturated hydraulic conductivities (-10 to -2 kPa, 0 to 20 cm depth) for this CT field compared to the NT field. In contrast, Miller et al. (1998) found few differences in biopores (root channels, earthworm channels) or cracks in these same fields.

It is interesting to note that our NT soils have a dramatically lower degree of water flow preference than NT soils in eastern North America (Granovsky et al. 1994). We attributed this difference to the absence of the "nightcrawler" earthworm *(Lumbricus terrestris)* in our NT fields. This earthworm forms deep (>50 cm), permanent and vertically oriented burrows that dramatically enhance preferential leaching of water and chemicals in NT fields of North America (Edwards et al. 1989). In contrast, our NT field contains a smaller earthworm, *Aporrectodea caliginosa* (Clapperton et al. 1997), which forms shallow (≤30 cm) and randomly oriented burrows (McKenzie and Dexter 1993). The latter earthworm's ecology is consistent with the lower degree of water flow preference found in our NT fields.

The stability of spatial outflow patterns was investigated by com-

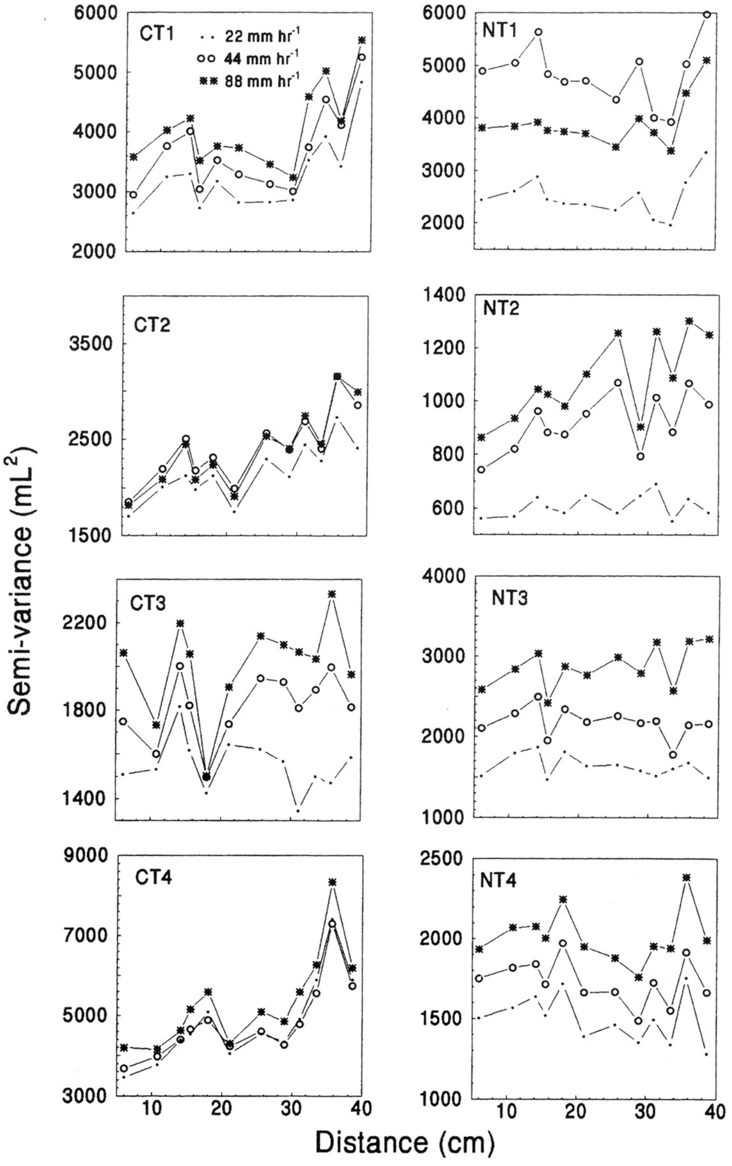

Fig. 4. Semi-variogram of cell water outflow (mL) from soil blocks as affected by tillage practice and rainfall intensity. Tillage treatments are conventional tillage (CT) and no-till (NT).

paring cell water outflows at the different rainfall intensities. A positive linear relationship was found between cell outflows for the three rainfall comparisons of 44 versus 22, 88 versus 22 and 88 versus 44 mm h^{-1} (Table 2). The R^2 values for all comparisons ranged from 0.79 to 0.97 and were all significant ($P < 0.001$). Slopes ranged from 0.87 to 1.36 and mean values

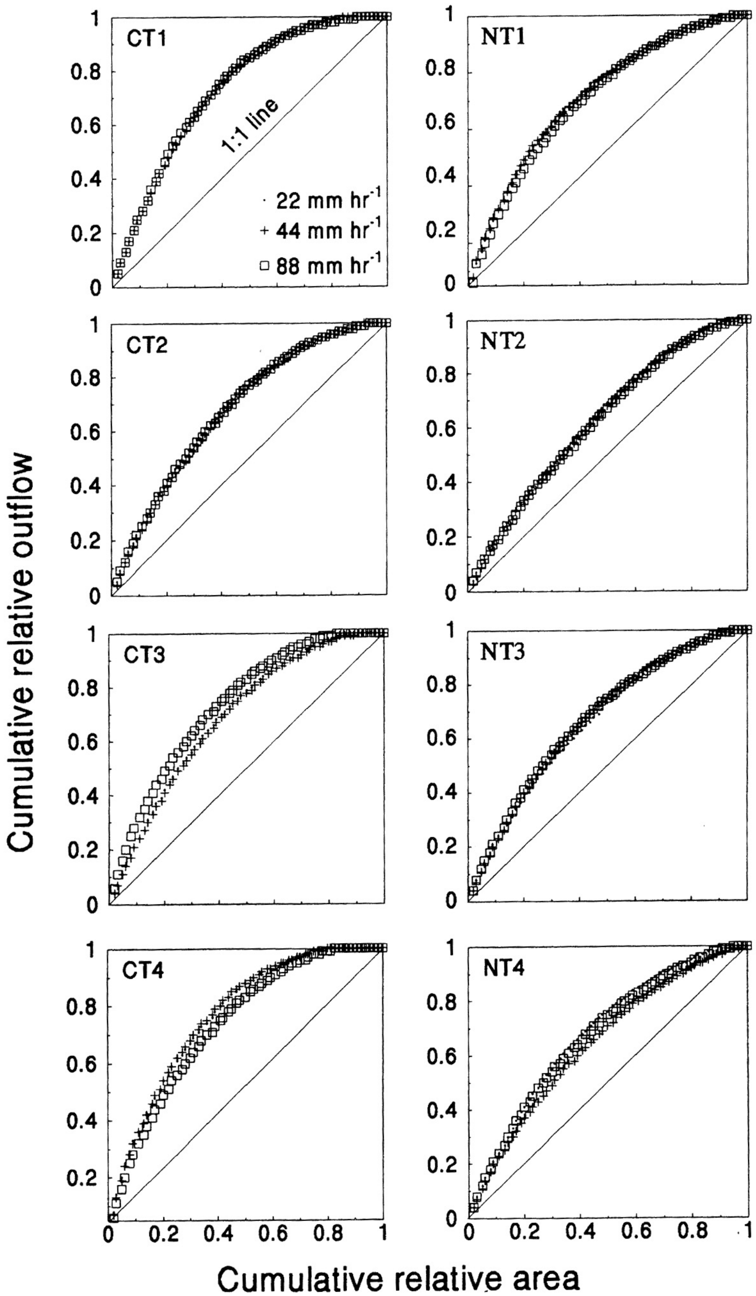

Fig. 5. Degree of water flow preference for soil blocks as affected by tillage practice and rainfall intensity. Tillage treatments are conventional tillage (CT) and no-till (NT).

Table 2. Regressions of rainfall intensity (mm h^{-1}) comparisons for cell water outflow (% of one pore volume) for each soil block

Tillage[b]	Rainfall intensity (mm h^{-1}) comparison[a]								
	44 (y) versus 22 (x)[2]			88 (y) versus 22 (x)			88 (y) versus 44 (x)		
	m	b	R^2	m	b	R^2	m	b	R^2
CT1	1.02	0.002	0.93***	1.02	0.009	0.82***	0.98	0.009	0.86***
CT2	1.03	0.013	0.95***	1.01	0.012	0.92***	0.98	-0.001	0.97***
CT3	0.97	0.016	0.82***	1.01	0.035	0.79***	1.03	0.019	0.95***
CT4	0.98	0.005	0.95***	1.01	0.032	0.92***	1.03	0.031	0.90***
Mean	1.00	0.009	0.91	1.01	0.022	0.86	1.00	0.015	0.92
SE	0.02	0.003	—	0.003	0.007	—	0.01	0.007	—
NT1	1.36	0.001	0.96***	1.19	0.016	0.90***	0.87	0.016	0.95***
NT2	1.18	0.012	0.93***	1.27	0.016	0.92***	1.07	0.005	0.97***
NT3	1.06	0.021	0.85***	1.24	0.016	0.88***	1.09	0.004	0.90***
NT4	1.03	0.020	0.91***	1.06	0.029	0.84***	1.02	0.010	0.90***
Mean	1.16	0.013	0.91	1.19	0.019	0.89	1.01	0.009	0.93
SE	0.07	0.005	—	0.05	0.003	—	0.05	0.003	—

[a]Slope (m), intercept (b) and coefficient of multiple determination (R^2) for regressions of rainfall intensity (mm h^{-1}) comparisons for cell water outflow (% of one pore volume) for each soil block. R^2 is significant at 0.05 (*), 0.01 (**) and 0.001 (***).

[b]CT, conventional tillage; NT, no-till.

were generally near one. These results indicated that the pattern of water outflow through soil pores was remarkably stable over time and among rainfall intensities. In contrast, Ogden et al. (1999) reported a decrease in r values (coefficient of correlation) for cell water outflow when the second to fifth rainfall events were compared to the first event. They applied 40 mm of water with a rainfall intensity of 38 mm h^{-1}, and the five successive rains were separated by intervals of 4, 8, 9 and 50 d, respectively. We attributed our stable flow pattern over time to the protective sand layer we used on top of each soil block, which prevents surface sealing of macropores (Ela et al. 1992). Ogden et al. (1999) did not use a protective sand layer, which likely contributed to a decrease in pore stability over time.

We found no tillage effect on the temporal changes in outflow patterns, even though NT soils in this area have a significantly higher percentage of water-stable aggregates (Dormaar and Lindwall 1989). Ogden et al. (1999) found that *r* values for NT cell outflow decreased considerably more than CT with successive rainfall events. They found the instability of NT drainage patterns was unexpected in a structurally stable soil. Maximum outflow in our soil blocks generally occurred in the same cell for each successive rainfall intensity, which also indicated relatively stable flow through the largest macropores. Maximum values occurred in the same cell for all three rainfall intensities for blocks CT2, CT4 and NT1; in the same cell for two of the three rainfall intensities for blocks CT3, NT2, NT3 and NT4; and in a different cell for each rainfall intensity for block CT1 (Fig. 2 and 3).

The number of visible pores (>500 µm diam.) on the bottom of the soil blocks ranged from 28 to 583 (132 to 2944 m^2) for CT and from 13 to 205 (60 to 1012 m^2) for NT. The mean number of pores was generally greater for CT (274±124) than NT (108±45), suggesting a greater potential for preferential flow under CT. In contrast, Logsdon et al. (1990) reported a greater number of pores (>400 µm diam.) in NT (290 to 460 m^2) fields compared to chisel plow CT (170 to 340^2) fields. Granovsky et al. (1993) also found a greater numbers of pores (>2,000 µm) in NT (44 to 400^2) soil blocks than moldboard plowed CT (0 to 211m^2) blocks. We found no positive linear relationship between cell outflow and the number of visible pores on the basal area of each cell (data not shown). At the block level, however, we found significant positive linear relationships between outflow and number of visible pores for the NT blocks at 22 and 44 mm h^{-1} (Fig. 6). Granovsky et al. (1994) also reported no correlation between cell outflow and the presence of macropores in NT soil blocks, but did observe a positive linear relationship between outflow and macropore area at the block level. They noted that about one-half of all macropores visible on the basal block area were hydraulically active. Macropore continuity cannot be derived from visible macropores on a two-dimensional soil surface (Singh and Kanwar 1991), which may explain the lack of correlation we observed at the cell level. In addition, water flow through soils depends on the complex interaction of many factors, including soil structure, texture, biopores and cracks (Logsdon et al. 1990).

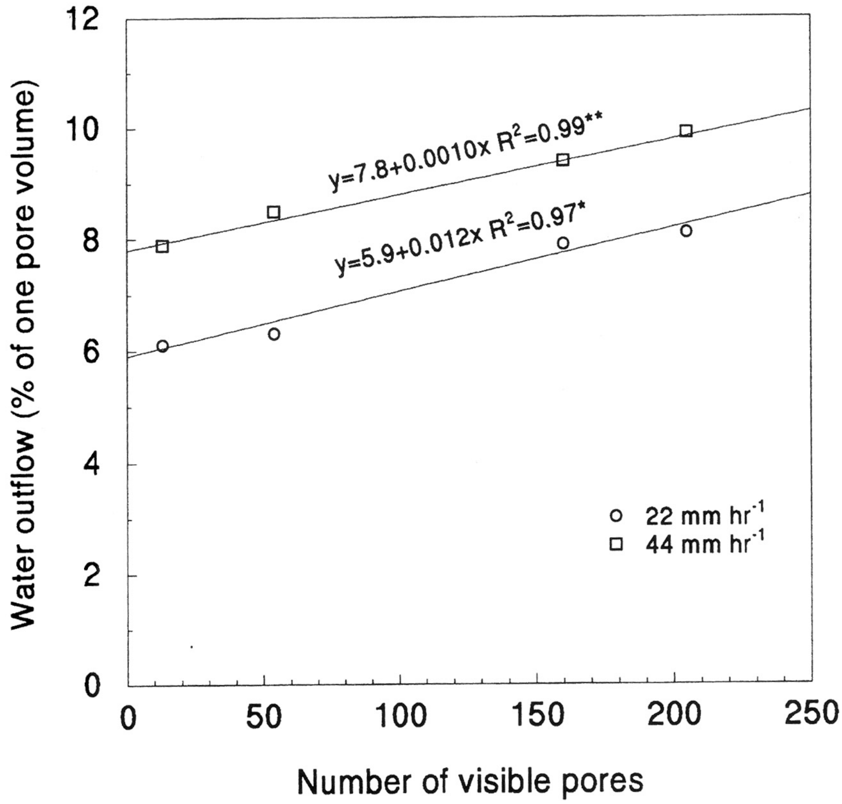

Fig. 6. Relationship between number of visible pores (>500 μm diam.) and water outflow (block level) for NT soil blocks at rainfall intensities of 22 and 44 mm h^{-1}.

Antecedent soil water was higher for NT than CT at all rainfall intensities (Table 1) and was consistent with surface ponding that occurred more frequently on the NT than CT blocks. However, regression analysis indicated there was no relationship (for each tillage and rainfall intensity) between antecedent soil water and the four water outflow parameters shown in Table 1 (data not shown). The only exception we found was for the NT blocks at the 88 mm h^{-1} intensity, where there was a negative linear relationship ($R^2 = 0.95$, $P < 0.05$) between antecedent water and total block outflow. Granovsky et al. (1994) reported that antecedent soil water was the dominant factor affecting transport of water in NT soil blocks; it was negatively correlated with the degree of water flow preference and positively correlated with total leachate volume. Other researchers have reported both positive (Seyfried and Rao 1987; Bouma 1991) and negative (Nielsen and Biggar 1961; White 1985) relationships between antecedent soil water and preferential flow. The contrasting findings may be due to the complex nature of preferential flow in unsaturated soils, which is dependent on the water content of the soil and the rate of application as well as lateral movement from larger water-conducting pores into the adjacent soil matrix (Bouma 1981).

Chloride Outflow

A total of 4430 mg of chloride tracer was applied to the top of each soil block. The spatial variation of the mass of chloride (mg) in outflow is shown on Fig. 7 and 8. Unlike water outflow, chloride outflow was detected in ≤14% (≤9 of 64 cells) of cells. This indicted that chloride outflow occurred through only a small percentage of the block area. The maximum concentration of chloride in a cell was as high as 30.13 mg L^{-1} (Fig. 9). Mean values for maximum chloride concentration were higher for CT (6.92± 2.25 mg L^{-1}) than NT (2.48±1.18 mg L^{-1}) at 88 mm h^{-1}, but not at the two lower intensities. Rainfall intensity had no effect on mean values for maximum chloride concentration for either CT or NT.

Most soil block studies have reported a positive relationship between rainfall intensity and preferential leaching of water and solutes (Edwards et al. 1992; Quisenberry et al. 1994). This has been attributed to the relationship between application rate and the infiltration rate of the soil matrix (Bouma 1991). At high rainfall intensities (application rate > infiltration rate) preferential flow may be almost instantaneous, and at very low intensities (application rate < infiltration rate) it may never occur (Bouma 1991). In addition, at extremely low application rates and when the soil is drier, lateral infiltration or diffusion of water and solutes from macropores into the drier soil matrix may also decrease preferential leaching (Bouma 1991). In contrast, some researchers have reported a negative relationship between rainfall intensity and preferential leaching of water and solutes (Phillips et al. 1995).

Total mass loss of chloride from each soil block ranged from 0 to 2.72 mg, which was <0.1% of the total amount applied. Since our detection limit for Cl^- was 2.13 mg L^{-1}, the question arises as to whether a significant mass of Cl^- leached through but was undetected because of the existing detection limit. If all 64 cells had a Cl^- concentration just below the detection limit (e.g., 2.00 mg L^{-1}), and using the maximum water outflow observed for a cell (385.47 mL), the mass of Cl^- for one cell would be 0.771 mg, and for the block (64 cells) it would be 49.3 mg. Since 4430 mg of Cl^- was applied to the block, the total potential mass loss of Cl^- from a block that would be undetected would be only 1.1% of the total amount applied. Therefore, low recovery of the applied Cl^- tracer in the leaching water suggests that most of the Cl^- still remained within the soil block. If anion exclusion dominated Cl^- leaching, most of the applied Cl^- would have been recovered in the leaching water. We believe that most of the water containing chloride infiltrated into the soil matrix (immobile water regions) and became unavailable for bypass flow through macropores. Infiltration into the soil matrix from the soil surface, lateral diffusion from macropores into the soil matrix and infiltration into discontinuous macropores (internal catchment) can all contribute to immobile or stagnant

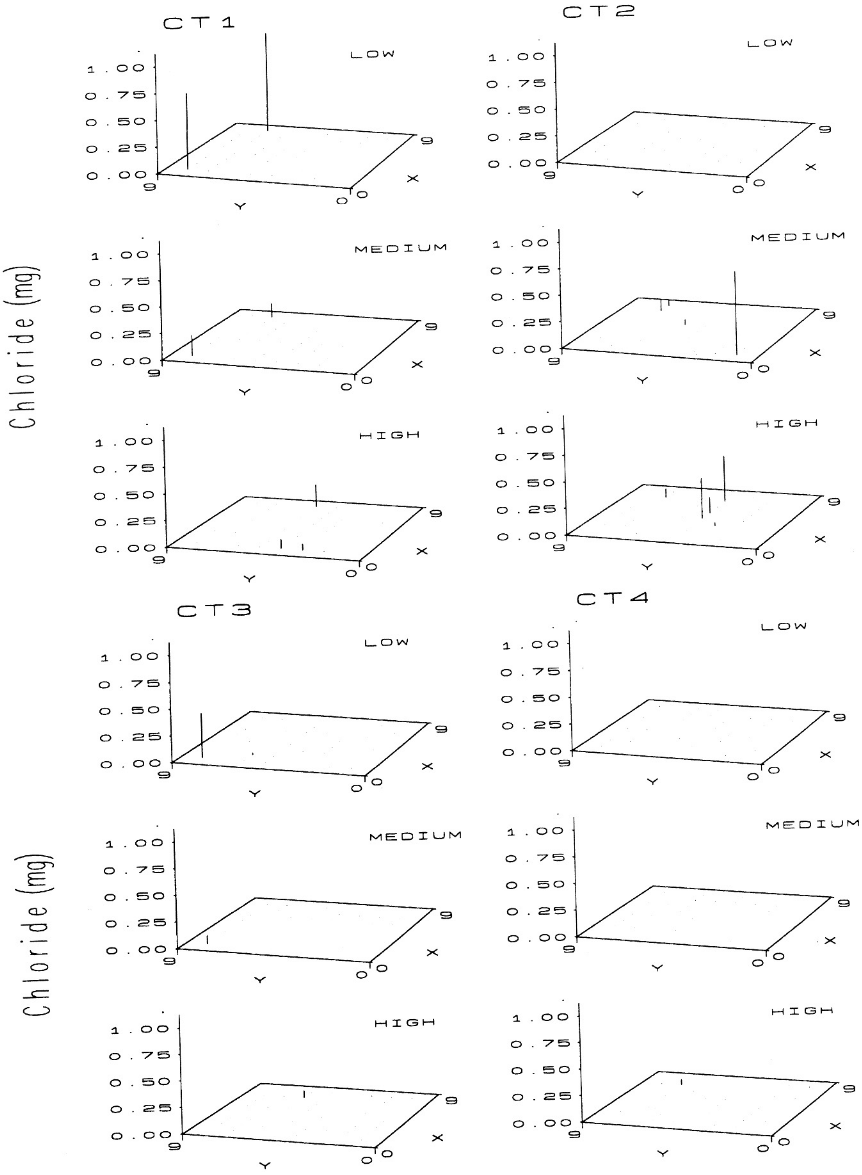

Fig. 7. Cell chloride outflow (mg) from soil blocks taken from conventional tillage (CT) field in southern Alberta. Rainfall intensity treatments are low, medium and high intensities (22, 44 and 88 mm h^{-1}, respectively).

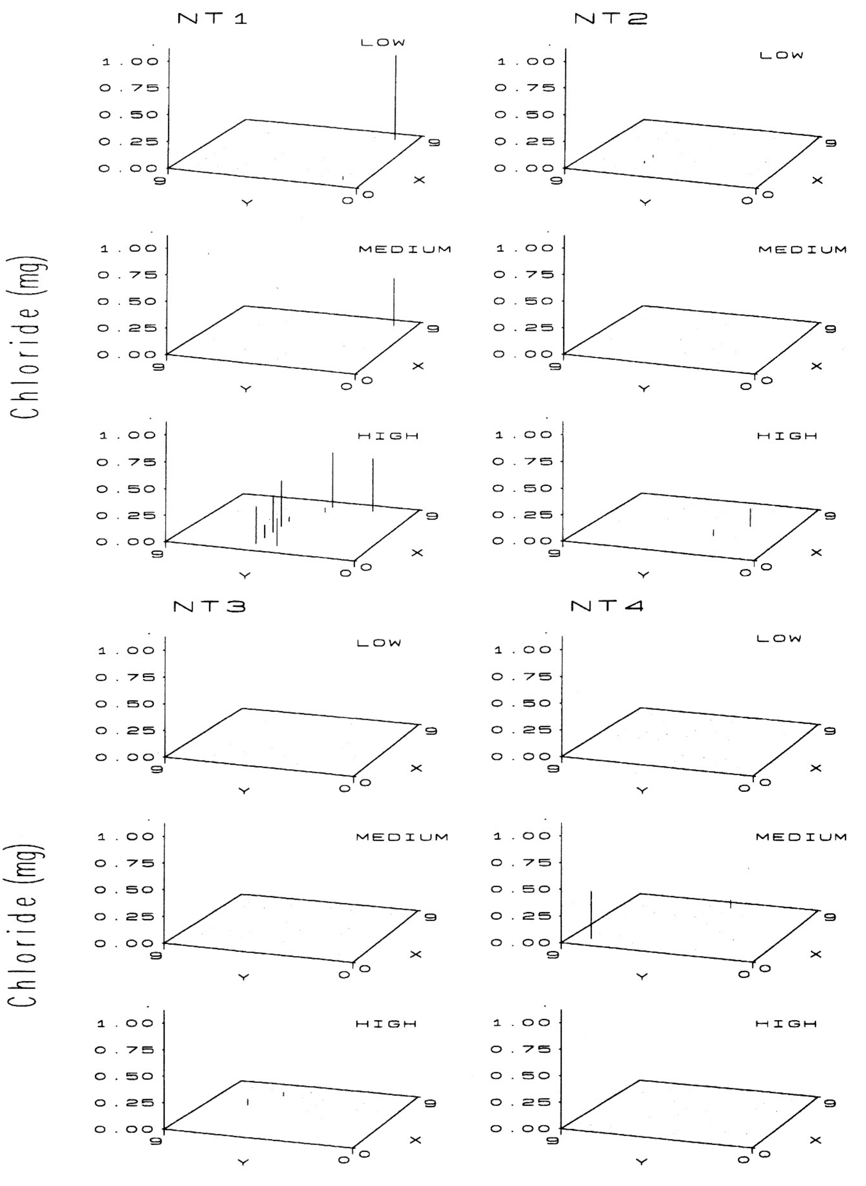

Fig. 8. Cell chloride outflow (mg) from soil blocks taken from no-till (NT) field in southern Alberta. Rainfall intensity treatments are low, medium and high intensities (22, 44 and 88 mm h^{-1}, respectively).

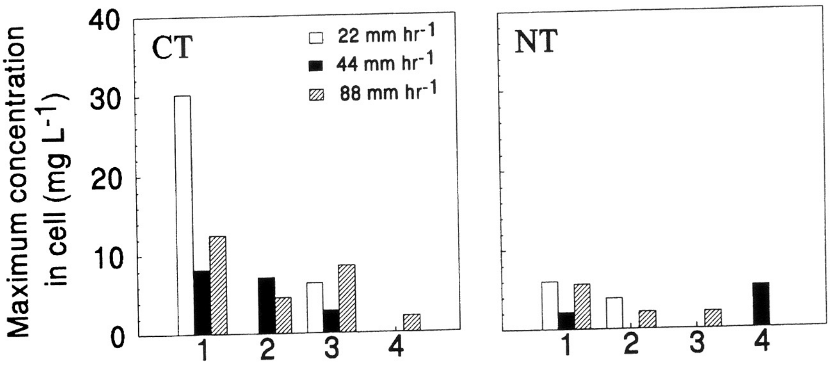

Fig. 9. Maximum concentration of chloride in a cell as affected by tillage practice and rainfall intensity. Tillage treatments are conventional tillage (CT) and no-till (NT).

regions of water (Bouma 1991). In contrast, Edwards et al. (1994) reported much higher amounts of bromide (up to 11% of total amount applied) that leached through their NT soil blocks.

We found no relationship between the mass of chloride and volume of water leached at either the cell or block level (data not shown). In contrast, other researchers have reported a good correlation between mass of chemical and volume of water in leachate at both the cell and block level (Edwards et al. 1992; Granovsky et al. 1994). Quisenberry et al. (1994) suggested that a good correlation between chloride and water outflow (cell level) indicated high preferential flow of chloride, and a poor correlation indicated low preferential flow of chloride.

It should be noted that the use of a protective sand layer on top of the soil block, application of the chloride tracer on the soil surface without incorporation, and gravity drainage on the bottom of the soil block all represent worst-case scenarios of preferential leaching for these conditions.

Summary and Conclusions

Water outflow through ≥98% of cells indicated that many soil pores contributed to leaching of water. The CV values for water outflow ranged from 44 to 105%, indicating a wide range in water outflow through small to large pores. Maximum cell outflow as a percentage of total block outflow ranged from 3.3 to 8.2% and was considerably less than reported elsewhere. Semi-variograms of cell water outflow indicated random flow, and the pattern of cell water outflow was stable with time and among rainfall intensities. The latter phenomenon was attributed to the presence of the protective sand layer on top of the soil block. Convex-shaped curves of cumulative water outflow from cells versus cumulative area indicated non-uniform water flow, but the degree of water flow preference was low. There were more visible macropores on the bottom of the CT than NT blocks, but there was no relationship between number of visible pores and water outflow at the cell or block level. The exception was for the NT blocks at 22 and 44 mm h^{-1}, where there was a significant positive linear relationship between water outflow and number of visible pores at the block level. We found no relationship between antecedent soil water content and degree of water flow preference for each tillage practice and rainfall intensity. The only exception was for the NT blocks at 88 mm h^{-1}, where there was a negative linear correlation.

Tillage practice had a stronger influence on water outflow than rainfall intensity. The mean CV values for water outflow were generally higher for CT than NT at 22 and 88 mm h^{-1}; the degree of water flow preference was higher for CT than NT at all three rainfall intensities; and the mean value for total water outflow (block level) was higher for CT than NT at 22 mm h^{-1}. Greater preferential water flow under CT than NT may be related to the greater rooting density under CT. Rainfall intensity had little or no influence on water outflow parameters. The exception was for the mean value of total water outflow (block level), which increased with greater rainfall intensity for CT and, to a lesser extent, for NT.

Chloride outflow in ≤14% of cells indicated that only a few soil pores contributed to chloride leaching. Total mass loss of chloride from each soil block was <0.1% of the total amount applied, which suggested most chloride was retained within the soil block, and was more consistent with immobile water rather than anion exclusion. Mean values of maximum chloride concentration were higher for CT than NT at 88 mm h^{-1}, but not at the two lower intensities. Rainfall intensity had no effect on mean values of maximum chloride concentration for either CT or NT.

Our results indicated that although preferential leaching occurred in these wet clay loam soils under relatively high rainfall intensities, the amount of chloride leached past 50 cm was minimal. Caution is advised in extrapolating these results to scenarios where conditions are different from this study.

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References

Andreini MS, Steenhuis TS. 1990. Preferential paths of flow under conventional and conservation tillage. Geoderma **46:**85–102.

Bevan K. 1991. Modelling preferential flow: an uncertain future, p. 1–13. *In* Gish TJ, Shirmohammadi A (ed.), Preferential flow. Proc. Nat. Symp., Dec. 16–17, Chicago.. Amer. Soc. Agric. Eng., St. Joseph, MI.

Bouma J. 1981. Soil morphology and preferential flow along macropores. Agric. Water Manage **3**:235–250.

Bouma J. 1991. Influence of soil macroporosity on environmental quality. Adv. Agron. **46**:1–34.

Bowman BT, Brunke RR, Reynolds WD, Wall GJ. 1994. Rainfall simulator-grid lysimeter system for solute transport studies using large, intact soil blocks. J. Environ. Qual. **23**:815–822.

Christiansen JE. 1942. Irrigation by sprinkling. Univ. of California Agric. Exp. Sta. Bull. 670.

Clapperton MJ, Miller JJ, Larney FJ, Lindwall CW. 1997. Earthworm populations as affected by long-term tillage practices in southern Alberta, Canada. Soil Biol. Biochem. **29**:631–633.

Dormaar JF, Lindwall CW. 1989. Chemical differences in Dark Brown Chernozemic Ap horizons under various conservation tillage systems. Can. J. Soil Sci. **69**:481–488.

Edwards WM, Shipitalo MJ, Owens LB, Norton LD. 1989. Water and nitrate movement in earthworm burrows within long-term no-till cornfields. J. Soil Water Conserv. **44**:240–243.

Edwards WM, Shipitalo MJ, Dick WA, Owens LB. 1992. Rainfall intensity effects transport of water and chemicals through macropores in no-till soils. Soil Sci. Soc. Am. J. **56**:52–58.

Edwards WM, Shipitalo MJ, Owens LB. 1993. Gas, water and solute transport in soils containing macropores: a review of methodology. Geoderma **57**:31–49.

Ela SD, Gupta SC, Rawls WJ. 1992. Macropore and surface seal interactions affecting water infiltration into soil. Soil Sci. Soc. Am. J. **56**:714–721.

Englund E, Sparks A. 1988. GEO-EAS (Geostatistical Environmental Assessment Software) user's guide. U.S. Environ. Protect. Agency, Las Vegas, NV.

Granovsky AV, McCoy EL, Dick WA, Shipitalo MJ, Edwards WM. 1993. Water and chemical transport through long-term no-till and plowed soils. Soil Sci. Soc. Am. J. **57**:1560–1567.

Granovsky AV, McCoy EL, Dick WA, Shipitalo MJ, Edwards WM. 1994. Impacts of antecedent moisture and soil surface mulch coverage on water and chemical transport through a no-till soil. Soil Tillage Res. **32**:223–236.

Jury WA, Gardner WR, Gardner WH. 1991. Soil physics. John Wiley and Sons, Inc., New York.

Kodama H. 1979. Clay minerals in Canadian soils: their origin, distribution and alteration. Can. J. Soil Sci. **59**: 37–58.

Logsdon SD, Allmaras RR, Wu L, Swan JB, Randall GW. 1990. Macroporosity and its relation to saturated hydraulic conductivity under different tillage systems. Soil Sci. Soc. Am. J. **54**:1096–1101.

Luxmoore RJ. 1991. On preferential flow and its measurement, p. 113–121. *In* Gish TJ, Shirmohammadi A (ed.), Preferential flow. Proc. Nat. Symp., Dec. 16–17, Chicago. Amer. Soc. Agric. Eng., St. Joseph, MI.

McKenzie BM, Dexter AR. 1993. Size and orientation of burrows made by the earthworms *Aporrectodea rosea* and *A. caliginosa*. Geoderma **56**:233–241.

Miller JJ, Kokko EG, Kozub GC. 1998. Comparison of porosity in a Chernozemic soil. Can. J. Soil Sci. **72**:569–579.

Miller JJ, Lamond BJ, Sweetland NJ, Larney FJ. 1999a. Preferential leaching in large undisturbed soil blocks from conventional tillage and no-till fields in southern Alberta. Water Qual. Res. J. Canada **34:**249–266.

Miller JJ, Larney FJ, Lindwall CW. 1999b. Physical properties of a Chernozemic clay loam soil under long-term conventional tillage and no-till. Can. J. Soil Sci. **79:**325–331.

Nielsen DR, Biggar JW. 1961. Miscible displacement in soils: I. Experimental information. Soil Sci. Soc. Am. J. **25**:1–4.

Ogden CB, van Es HM, Wagenet RJ, Steenhuis TS. 1999. Spatial-temporal variability of preferential flow in a clay soil under no-till and plow-till. J. Environ. Qual. **28:**1264–1273.

Phillips RE, Quisenberry VL, Zeleznik JM. 1995. Water and solute movement in an undisturbed, macroporous column: extraction pressure effects. Soil Sci. Soc. Am. J. **59:**707–712.

Quisenberry VL, Phillips RE, Zeleznik JM. 1994. Spatial distribution of water and chloride macropore flow in a well-structured soil. Soil Sci. Soc. Am. J. **58:**1294–1300.

Schafer WM and Singer MJ. 1976. A new method of measuring shrink-swell potential using soil pastes. Soil Sci. Soc. Am. J. **40**: 805–806.

Seyfried MS, Rao PSC. 1987. Solute transport in undisturbed columns of an aggregated tropical soil: preferential flow effects. Soil Sci. Soc. Am. J. 51:1434–1444.

Shipitalo MJ, Edwards WM, Dick WA, Owens LB. 1990. Initial storm effects on macropore transport of surface-applied chemicals in no-till soil. Soil Sci. Soc. Am. J. **54:**1530–1536.

Singh P, Kanwar RS. 1991. Preferential solute transport through macropores in large undisturbed saturated soil columns. J. Environ. Qual. **20:**295–300.

Tautchin M, Dzikowski P, Chetner S. 1991. Intensity, duration and frequency of extreme rainfall in Alberta. Conservation and Development Branch, Alberta Agriculture, Food and Rural Development, Edmonton, AB.

Technicon. 1974. Chloride in water and wastewater. Industrial Method No. 99-70W/B. Technicon Industrial Systems, Tarrytown, NY.

Topp GC. 1993. Soil water content, p. 541–558. *In* Carter MR (ed.), Soil sampling and methods of analysis. Lewis Publ., Boca Raton, FL.

Tossell RW, Dickinson WT, Rudra RP, Wall GJ. 1987. A portable rainfall simulator. Can. Agric. Eng. **29:**155–162.

Volkmar KM, Entz T. 1995. A method for characterizing the effect of root-formed pores on growth of roots in a Chernozemic clay loam. Can J. Soil Sci. **75:**293–298.

White RE. 1985. The influence of macropores on the transport of dissolved and suspended matter through soil. Adv. Soil Sci. **3**:95–119.

Wildenschild D, Jensen KH, Villholth K, Illangasekare TH. 1994. A laboratory analysis of the effect of macropores on solute transport. Ground Water **32:**381–389.